LLL-L-51D
September 4, 1985
SUPERSEDING
LLL-L-51C
November 3, 1972

FEDERAL SPECIFICATION

LADDERS, PORTABLE, WOOD: EXTENSION AND STRAIGHT

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wooden ladders, both single and extension.

1.2 Classification.

1.2.1 Classes, types, and lengths. The ladders shall be of the following classes, types, and lengths, as specified (see 6.2).

Class A - Single ladder.
Type IA - 300 lb. duty rating
Type I - 250 lb. duty rating
Type II - 225 lb. duty rating
Type III - 200 lb. duty rating
Length - 8 feet through 30 feet by intervals of 2 feet.

Class B - Extension ladder, two sections.
Type IA - 300 lb. duty rating
Type I - 250 lb. duty rating
Type II - 225 lb. duty rating
Type III - 200 lb. duty rating
Length - 20 feet through 60 feet by intervals of 4 feet.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of the invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver,CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.
MIL-C-11133 - Crates; Shipping, Wood, Open, Wirebound.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Underwriters Laboratories, Inc., (UL) Standard

UL 112 - Portable Wood Ladders

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, NW, Washington, DC 20036).

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606)

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3. REQUIREMENTS

3.1 Description. The ladders shall conform to UL STD 112.

3.2 Nonslip bases. When specified (see 6.2) the ladders shall be equipped with nonslip bases.

3.3 First article. The supplier shall furnish one or more sample ladders as specified in the solicitation/contract for examination and test as specified by this document, including testing for compliance with UL STD 112.

3.4 Casualty hazard. Each contractor shall submit to the contracting agency proof that the ladders they propose to supply under this specification conform to the requirements of UL STD 112. The label, or listing mark of the UL, may be accepted as evidence that the ladders conform to this paragraph.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.5)

4.3 First article ladder inspection.

4.3.1 Examination. The first-produced ladders shall be examined for the visual dimensional requirements of UL STD 112. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article ladders shall be tested as specified in UL STD 112. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of all ladders of the same class, type, and size submitted for inspection at one time.

4.4.2 Sampling.

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4.4.2.1 Sampling for visual and dimensional requirements. Sampling shall be in accordance with MIL-STD-105, inspection level II. AQL shall be 4.0 percent defective.

4.4.2.2 Sampling for test requirements. Sampling shall be in accordance with MIL-STD-105, inspection level S-2. AQL shall be 4.0 percent defective.

4.4.3 Examination.

4.4.3.1 Individual. Each type and size of ladder shall be examined for the visual dimensional requirements of UL STD 112. The presence of a defect shall be cause for rejection.

4.4.4 Test samples.

4.4.4.1 Samples selected in accordance with 4.4.2.2 shall be tested as specified in UL STD 112. Failure of any test shall be cause for rejection.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

1. Extension ladders not retracted and secured as specified for level A.
2. Rope not removed, packaged, and secured as specified for level A.
3. Crate not as specified for levels A or B.
4. Ladders packed together exceed crate limitations for levels A or B.
5. Shroud not provided for level A.
6. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be levels A or C as specified (see 6.2).

5.1.1 Level A. Extension ladders shall be retracted to minimum length and secured to prevent movement. The rope shall be removed and packaged in a waterproof bag conforming to MIL-B-117, type I, class B, style 1. The packaged rope shall be secured to the ladder by stapling beyond the heat seal in a manner to prevent destroying the waterproof nature of the bag.

5.1.2 Level B. The ladders shall be packed as specified in 5.1.1 for level A except a shroud shall not be required.

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5.1.2 Level C (Commercial packaging). The ladders shall be packaged in a manner to assure protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be at levels A, B, or C as specified (see 6.2).

5.2.1 Level A. The ladders shall be packed in close-fitting crate conforming to MIL-C-11133, type II, class 1 or 2. Ladders shall be packed individually or ladders of like description shall be packed together in quantities not to exceed the weight or dimension limitations of the crate. The weight or dimensional limitations of the crate shall be waived when the weight or dimensions of an individual ladder exceeds that limitation. The ladder shall be blocked, braced, anchored, and protected with a shroud in accordance with the appendix to the crate specification.

5.2.2 Level B. The ladders shall be packed as specified in 5.2.1 for level A except a shroud shall not be required.

5.2.3 Level C (Commercial packing). The ladders shall be packed to assure carrier acceptance and safe delivery to destination at the lowest transportation rates in compliance with the Uniform Freight Classification rules and National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military agencies. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. Wood ladders are intended to be used where electrical safety hazards may exist and metal ladders cannot be used.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Class, type, and length of ladder required (see 1.2).
(c) When nonslip bases are required.
(d) Level of packaging and level of packing required (see 5.1).

PREPARING ACTIVITY:

GSA - FSS

MILITARY INTERESTS:

Coordinating Activity

Army - ME

Review Activity

Army - ME

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